Proposed Solution

Wandr AI – AI Travel Itinerary Planner

1. Solution Overview

Wandr AI is a full-stack web application that accepts structured trip parameters from an authenticated user, forwards them to the Groq AI API (model: openai/gpt-oss-120b), parses the returned JSON itinerary, persists it in Supabase PostgreSQL, and renders it in a React frontend with editing, regeneration, and budget-tracking capabilities.

2. Core Components

Component	Technology	Responsibility
Landing Page	React + Tailwind	Communicate value proposition, drive signup
Auth Flow	Supabase Auth JS	Email/password signup & login, session management
Trip Creation Form	React Hook Form + Zod	Collect and validate trip parameters
API Client	TypeScript fetch	Attach Bearer JWT, call backend endpoints
FastAPI Backend	Python 3.12	Route handling, auth, orchestration
AI Service	Groq SDK + tenacity	Prompt construction, AI call, JSON parsing, retry
Itinerary Service	Supabase Python	CRUD operations on trips table
Budget Service	Pure Python	Cost aggregation and breakdown calculation
Database	Supabase PostgreSQL	Persistent storage with RLS

3. User Workflow

1. User visits landing page and clicks Get Started.
2. User signs up / logs in → Supabase issues JWT.
3. User opens New Trip form, fills all parameters, submits.
4. Frontend POST /api/trips/generate with Bearer JWT.
5. Backend verifies JWT, calls Groq, parses response, saves to Supabase.
6. Trip Detail page renders day-by-day cards.
7. User may regenerate any day or view the Budget tab.
8. Trip is saved; user returns to Dashboard or Trips list.

4. AI Prompt Strategy

The backend builds a structured prompt containing trip metadata (destination, dates, budget, travel style, interests, pace, dietary and accessibility needs). The system message instructs the model to return only valid JSON matching a predefined schema with days[], activities[], costs, accommodation suggestions, dining recommendations, and transportation notes. Temperature is set to 0.7 and max_tokens to 8000.